IGNITRON

AI AGENT FOR REAL-TIME SUGARCANE QUALITY INTELLIGENCE

ET THE ECONOMIC TIMES
AI HACKATHON 2026
THINK AI. BUILD WITH GENAI.

Problem Statement 5: Domain-Specialized AI Agents (Agriculture Domain)

TEAM MEMBERS

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PROBLEM AND MOTIVATION

Lack of real-time, non-destructive quality intelligence in sugarcane processing.

Problem

- Conventional sugarcane quality assessment relies on **destructive sampling and delayed testing**
- Evaluation is performed **post-crushing, not on intact cane flow**
- Random sampling leads to inaccurate and inconsistent quality estimation
- Lack of **real-time intelligence** results in inefficient mill decisions

Motivation

- Need for **non-destructive, real-time quality assessment**
- Improve **sugar yield and operational efficiency**
- Improve **sugar yield and operational efficiency**
- Ensure **transparent and fair pricing** for farmers

Fig 1. Problem and Motivation for Real-Time Non-Destructive Sugarcane Quality Monitoring.

To address these challenges, we propose an AI-driven multimodal sugarcane quality intelligence system.

BRIEF IDEA

- **Integrated real-time, non-destructive sugarcane quality monitoring system** using IR reflectance and dielectric/capacitive sensing with in-line multimodal data acquisition
- **AI-driven Brix prediction framework (refractometer-calibrated)** enabling accurate, continuous estimation without juice extraction (zero-loss measurement)
- **Intelligent, scalable decision-support platform** for quality classification and industry-ready deployment in conveyor-based mill operations

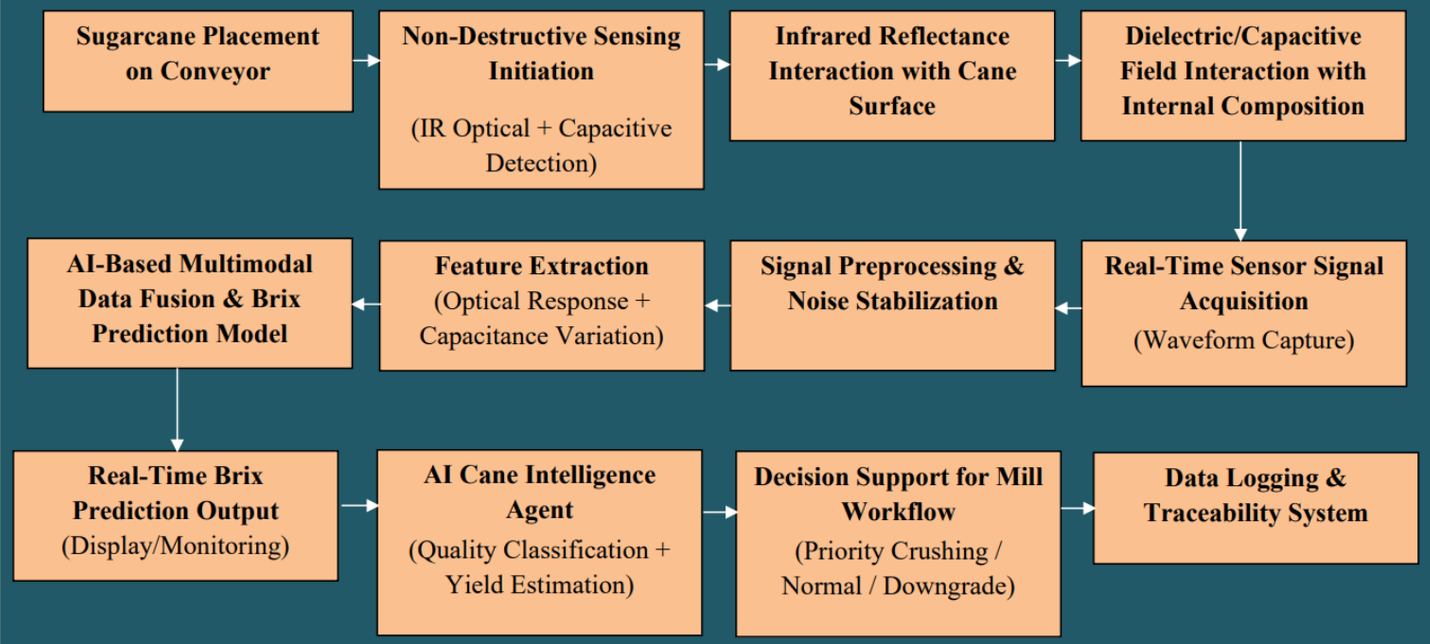

Fig 2. Process Flow of the AI-Driven Multimodal Sugarcane Quality Intelligence System.

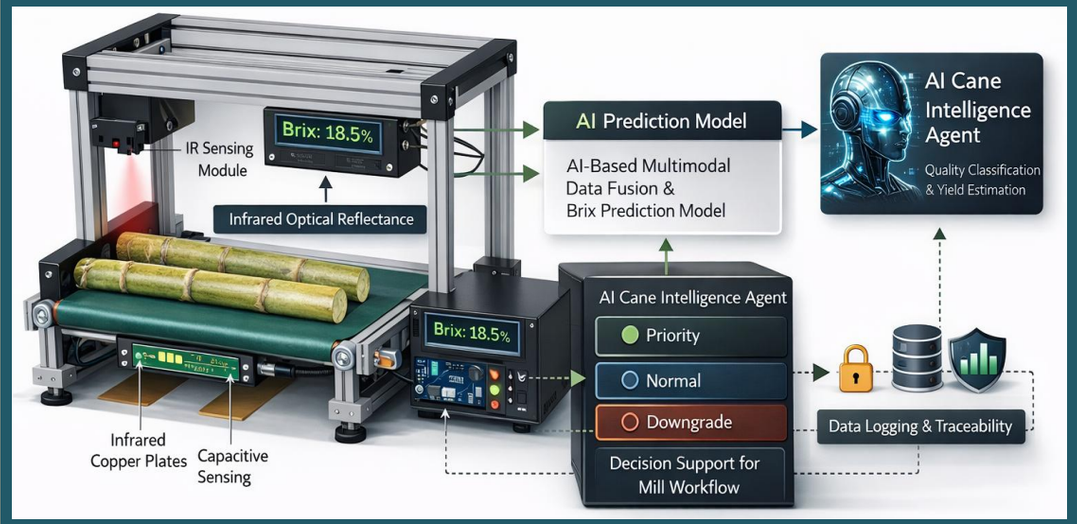

Fig 3. AI-Driven Multimodal Sugarcane Quality Intelligence System with Decision Support and Traceability.

KEY FEATURES AND TECHNOLOGIES

Key Features

- **Non-destructive in-line Brix estimation** directly on moving cane
- **Conveyor-integrated sensing module** for continuous operation
- **Multimodal sensing framework** combining optical and dielectric signals
- **AI-based prediction model** for accurate Brix estimation
- AI agent for quality classification and decision support
- Batch-wise data logging and traceability system

Technologies Used

- **Infrared Optical Reflectance Sensing** for surface interaction analysis
- **Dielectric/Capacitive Sensing** for internal composition detection
- **Multimodal Data Fusion Techniques** for robust prediction
- Real-time Signal Processing & Feature Extraction
- Embedded System (Arduino-based) for edge computation
- Industrial conveyor-integrated architecture

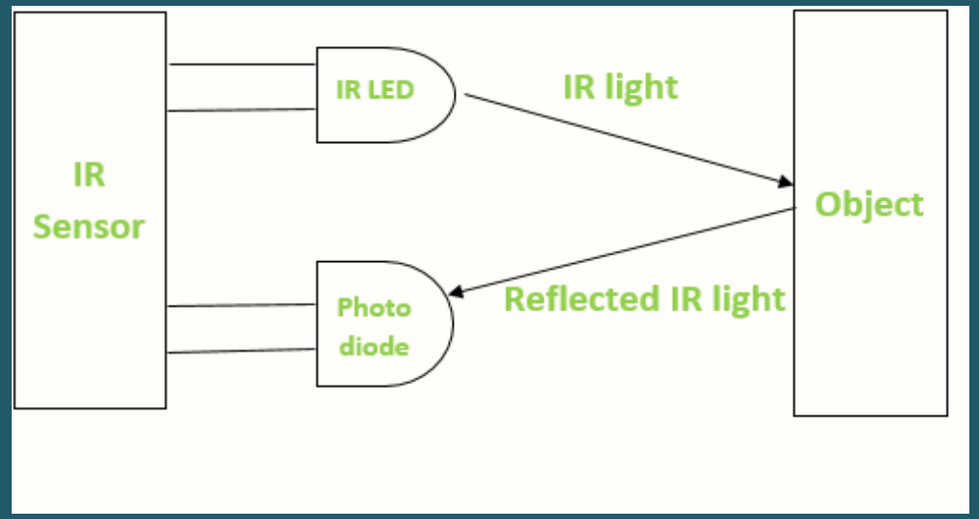

Fig.4: Optical IR Sensing Mechanism Using LED–Photodiode Reflectance.

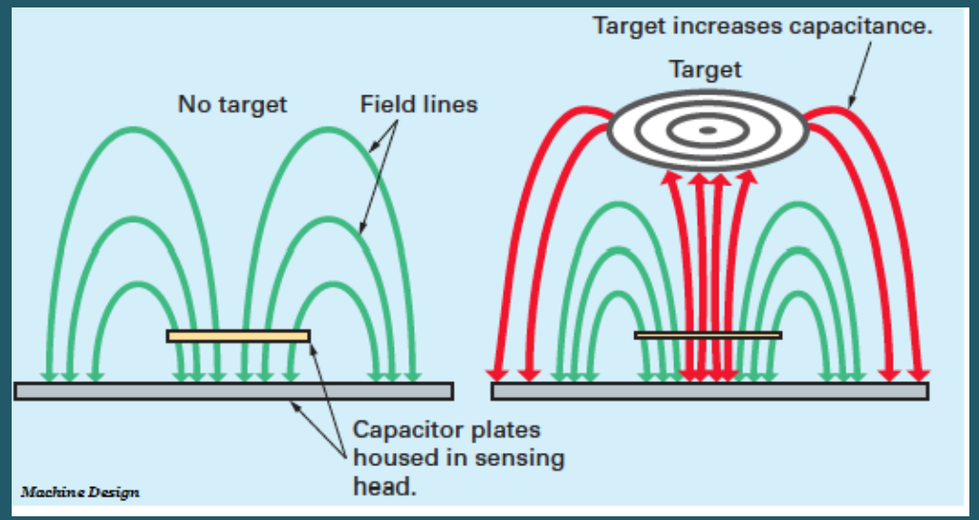

Fig.5: Dielectric/Capacitive Sensing Mechanism for Bulk Material Property Analysis.

CAD MODELS

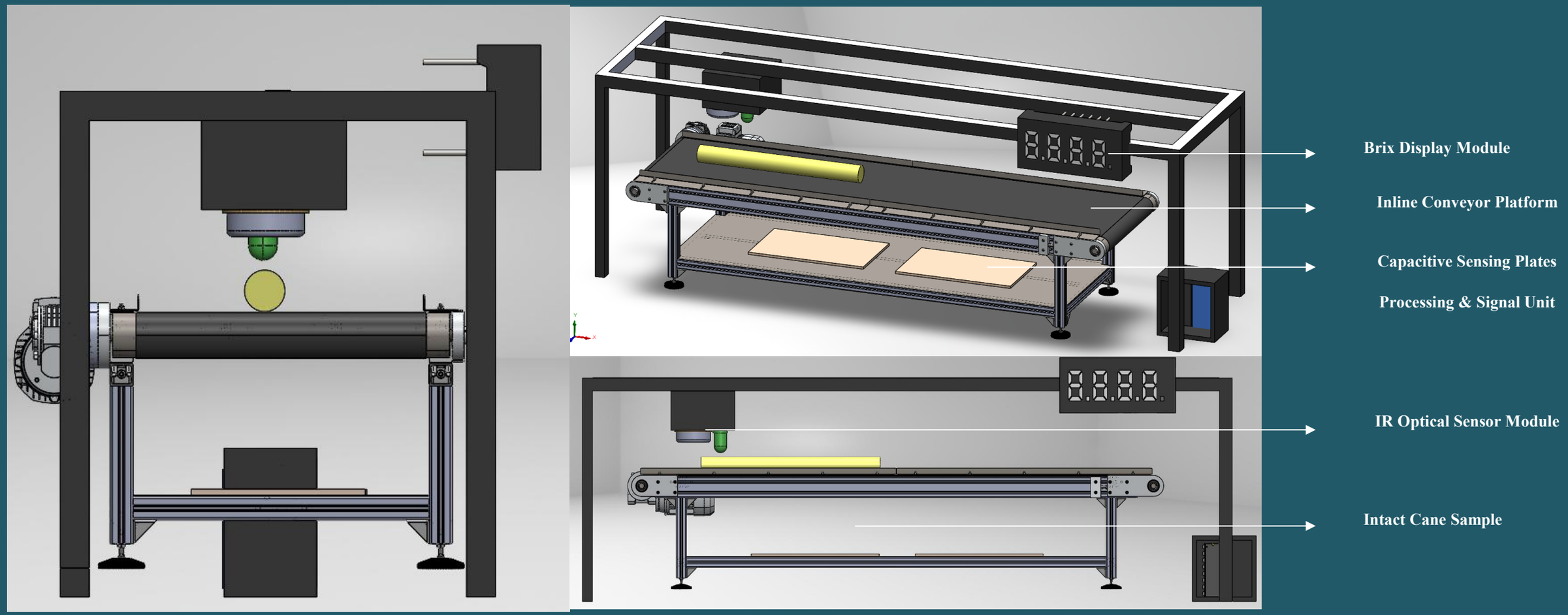

Fig.6: CAD Model of Conveyor-Integrated Multimodal Sugarcane Quality Monitoring System with IR and Capacitive Sensing Modules Designed in SolidWorks.

SYSTEM ARCHITECTURE OF AI-DRIVEN SUGARCANE QUALITY MONITORING SYSTEM

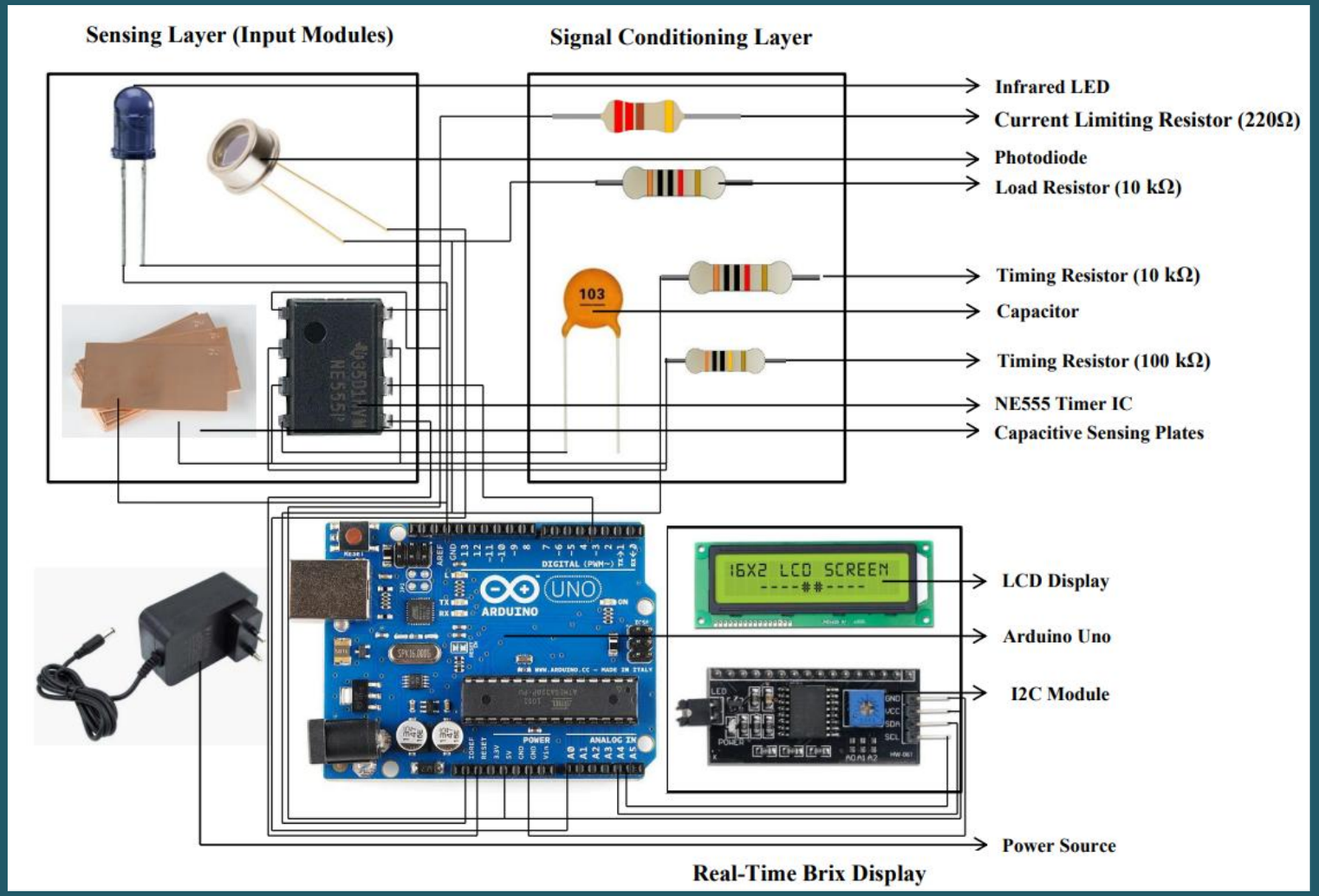

Fig.7: System Architecture of Multimodal AI-Based Sugarcane Quality Monitoring and Real-Time Brix Estimation.

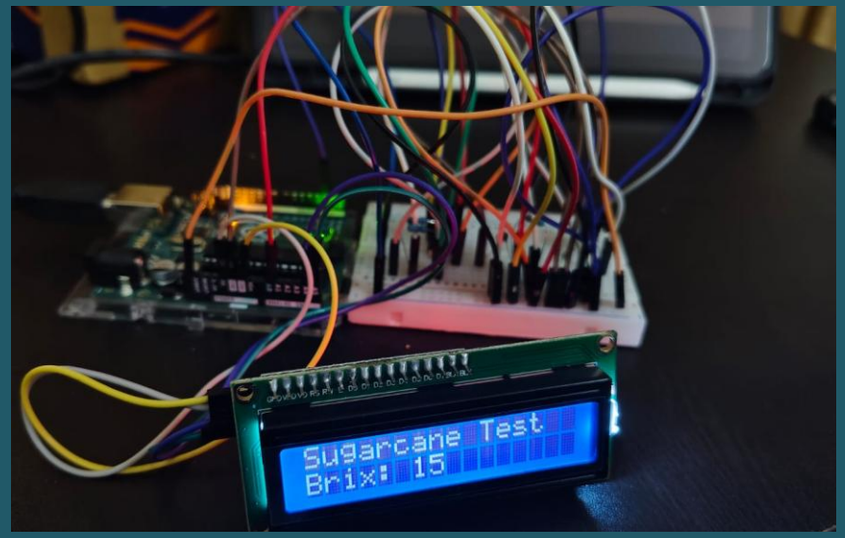

Fig.8: Hardware Prototype for Real-Time Sugarcane Brix Measurement.

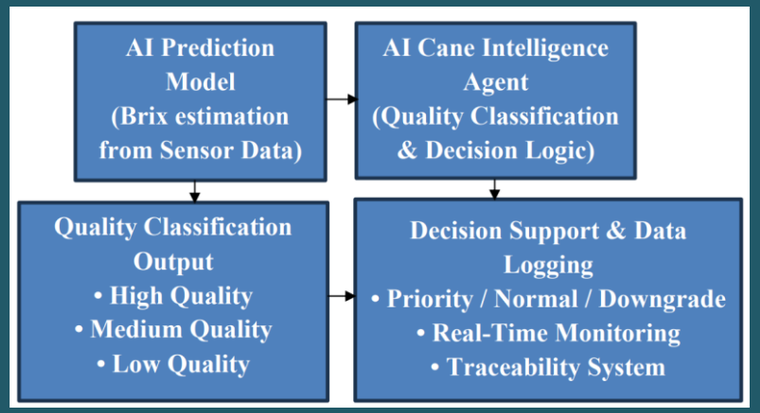

Fig.9: AI-Based Dashboard for Sugarcane Quality Monitoring and Decision Support.

AI-BASED REAL-TIME SUGARCANE QUALITY MONITORING SYSTEM

- **Real-time multimodal sensing prototype (IR + dielectric)** capturing in-line cane parameters and transmitting live data to the processing unit
- **Experimentally calibrated AI model** converting sensor inputs into accurate, real-time Brix values displayed on the dashboard interface
- **AI-based decision engine** providing instant quality classification (ACCEPT/REJECT/HOLD) with integrated monitoring and traceability

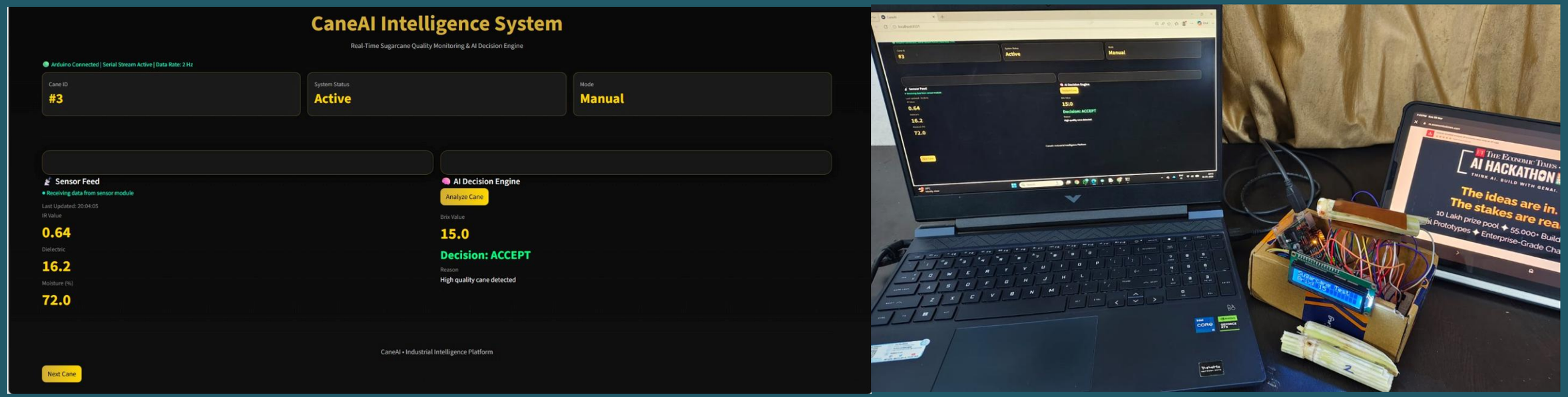

Fig.10: Prototype and dashboard for real-time Brix prediction and quality classification.

ESTIMATED IMPLEMENTATION COST

Table1: Bill of Materials and Cost Analysis for Sugarcane Quality Monitoring System.

Component / Module	Actual Item Used	Qty	Cost (₹)
Infrared Optical Sensing Unit	MAX30101 Photodetector Breakout	1	300
Embedded Processing Unit	Arduino Uno Rev3	1	1526
Display & Monitoring Interface	LCD1602 Display (I2C)	1	150
Signal Processing Circuit	LM358 Op-Amp	1	17
Timing / Control Support	NE555 Timer IC	1	10
Passive Components (Capacitors)	0.01μF Ceramic Capacitors	2	1
Passive Components (Resistors)	10kΩ + 100kΩ + 220Ω Resistors	3	2
Other miscellaneous items			200
Total Cost			2206

System Level	Estimated Cost
Prototype (Student Build)	₹1,000 – ₹3,000
Pilot Industrial Prototype	₹50,000 – ₹1,50,000
Full Industrial Deployment (Scaled System)	₹2,00,000 – ₹6,00,000

GitHub link: https://github.com/Anu-310/CaneAI-Quality-System/tree/main